



Reconstruction of *Microraptor* and the Evolution of Iridescent Plumage

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and 260 km (figs. S1 to S4). The overall distribution and thickness of proximal ejecta materials differed by a factor of ~ 3 , depending on resolution, which is small in comparison to the uncertainty in the magnetic paleofield strength. For impact angles of 30° from vertical, the projectile core materials were deposited in the central portion of the basin, where they can sink through the melt sheet. For impact angles of 60° from vertical, a larger fraction of the projectile escaped the Moon's gravity, and the projectile core materials were deposited exterior to the basin rim. For homogeneous projectiles (figs. S5 to S8), the projectile materials were deposited farther downrange than for a similar impact of a differentiated projectile. If the projectile materials had the magnetic properties of average chondritic meteorites, dipole field strengths of 100 μT would generate magnetic anomalies that are similar to those observed on the Moon. Larger impact velocities favor projectile vaporization, leading to weaker magnetic anomalies. Although both differentiated and undifferentiated projectiles can account for the distribution and intensities of lunar magnetic anomalies, differentiated projectiles with impact angles of 45° most easily account for the strong anomalies that are located near the rim of the South Pole–Aitken basin. In our simulations, some projectile materials were deposited far from the basin rim, and this distal ejecta could potentially explain the existence of strong isolated anomalies on the lunar nearside, such as Reiner- γ and Descartes (Fig. 1).

Large impact events were common in the early evolution of the solar system, and these would certainly have accreted important quantities of highly magnetic materials to the crusts of all the terrestrial planets and moons. A giant

northern lowlands-forming oblique impact on Mars (30, 31) could help to explain the existence of strong crustal magnetic anomalies in the southern highlands of Mars that are otherwise difficult to understand (32, 33). Similar magnetic anomalies might be expected to surround the Caloris basin on Mercury. Impact basin-associated magnetic anomalies should scale with the amount of retained projectile materials, and hence with basin size. Being exogenic in origin, planetary magnetic anomalies could be used to search for ancient meteoritic materials.

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Supporting Online Material

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SOM Text

Figs. S1 to S8

Tables S1 to S4

References (35–86)

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Reconstruction of *Microraptor* and the Evolution of Iridescent Plumage

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Iridescent feather colors involved in displays of many extant birds are produced by nanoscale arrays of melanin-containing organelles (melanosomes). Data relevant to the evolution of these colors and the properties of melanosomes involved in their generation have been limited. A data set sampling variables of extant avian melanosomes reveals that those forming most iridescent arrays are distinctly narrow. Quantitative comparison of these data with melanosome imprints densely sampled from a previously unknown specimen of the Early Cretaceous feathered *Microraptor* predicts that its plumage was predominantly iridescent. The capacity for simple iridescent arrays is thus minimally inferred in paravian dinosaurs. This finding and estimation of *Microraptor* feathering consistent with an ornamental function for the tail suggest a centrality for signaling in early evolution of plumage and feather color.

Feather colors in extant birds (Aves) are generated from pigments and a variety of nanostructural architectures (1, 2). Iridescent colors, an integral part of the avian plumage color gamut involved in signaling and display, are produced through coherent light scattering by laminar or

crystal-like arrays generated by layers of materials with different refractive indices—namely, keratin, melanin, and sometimes air—in feather barbules (1, 2). Melanosomes can be arranged in single or multiple layers (1, 2), and recent work shows that even slight organization of melanosomes can

produce weakly iridescent (glossy) colors (3). Iridescent nanostructures are diverse and have evolved independently numerous times in extant birds (4), but whether they are exclusively avian innovations or appear earlier in dinosaur evolution has been unknown.

Thus far, fossil evidence of iridescent plumage has been limited to a 47-million-year-old isolated avian feather from Germany (Grube Messel) (5). This feather preserved in fine nanostructural detail the organization typical of many iridescent avian melanosome arrays. Such pristine preservation is rare, however, and so far unknown in

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the 150- to 120-million-year-old fossils from China that have proven critical for investigating the relative timing of innovations in coloration and plumage evolution in dinosaurs (6, 7). Methods to investigate the presence of iridescent arrays in these and similarly preserved fossils have not been developed.

We investigated the relations between melanosome properties and iridescent arrays in Aves prompted by the melanosome characteristics and plumage preserved in a previously unknown fossil feathered dinosaur specimen (BMNHC PH881, Beijing Museum of Natural History; Fig. 1 and figs. S3 to S5) referred to as *Microraptor* (8). *Microraptor* has consistently been recovered as part of Paraves, a clade that includes troodontid, dromaeosaurid, and avialan theropod dinosaurs (9–11). The articulated specimen was recovered near Lamadong (Jianchang County, western Liaoning Province) and displays previously reported characteristics of this “four-winged” dinosaur [e.g., (9)], including asymmetrically vaned hind limb feathers that are about 80% of the length of the asymmetrically vaned primary wing feathers (Fig. 1 and fig. S4), as well as previously unknown plumage data. Feathers in BMNHC PH881 are particularly well preserved on the right forelimb, left hindlimb, and the tail as dark imprints (Fig. 1). No lighter and darker banding, spangles, or patches (6, 7, 12) were observed. Initial sampling of these feathers revealed apparent sheetlike, end-to-end

orientation of elongate melanosomes (Fig. 2F) similar to the subtle organization resulting in glossy black avian feathers (3).

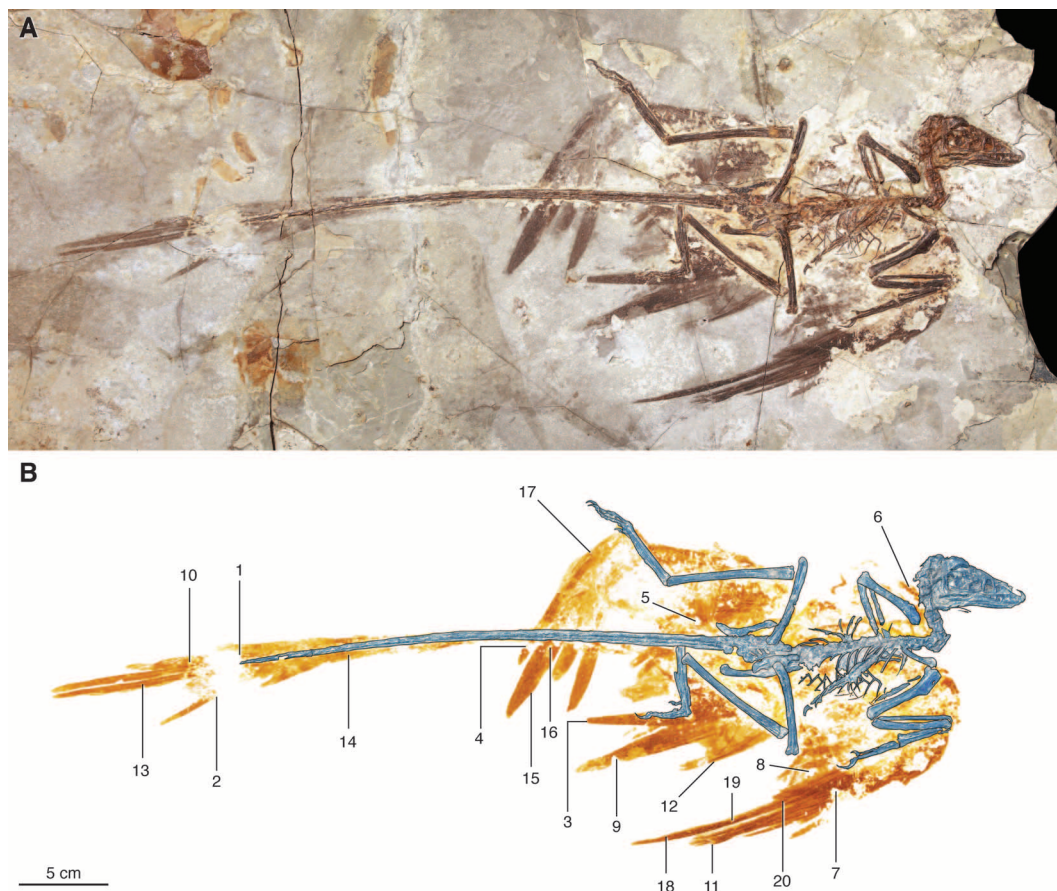
To investigate melanosome characteristics in avian iridescent arrays, we sampled melanosome morphology from a phylogenetically diverse set of iridescent feathers from extant birds (7, 8, 12) (table S4). Although some iridescent feathers contain highly divergent morphologies (e.g., hollow and flattened melanosomes in iridescent hummingbird feathers) that would be readily recognizable in the fossil record, most [13 out of 19 array types in (2)] contain rods similar to those in matte feathers. We focused on the latter in our sampling. These samples and denser sets from matte black, brown, and gray extant bird feathers were added to our existing database [(7, 12), new $n = 168$]. Relative to those from black feathers, melanosomes from iridescent feathers were found to be significantly longer (~1160 versus 1000 nm) and narrower (~211 versus 279 nm), providing higher aspect ratios (~5.4 versus 3.5; Figs. 2 and 3). Quadratic discriminant analysis (8), a standard method that allows classification of unknown samples from data on known samples, of these data predicted color of extant bird feathers with 82% accuracy (Fig. 2B and tables S1 and S2), showing that iridescent colors could reliably be detected. Samples of the 47-million-year-old fossil feather previously inferred as iridescent on the basis of nanoscale melanosome organization (5)

were also classified as iridescent with an 86% probability (Fig. 2B).

To then assess the colors and color patterns of the specimen, we took 26 samples of ~1 mm by 1 mm from throughout the feathered areas. All but six revealed numerous dense assemblages of melanosome imprints (8). Morphological measurements from melanosome imprints were acquired and analyzed in the same manner as the modern feathers (8). Melanosomes from BMNHC PH881 are within the range of extant melanosome dimensions and are predicted as either iridescent or black (Fig. 2 and table S2) with 58 to 100% probabilities. Six samples lacked melanosomes but had no associated light banding of the type correlated with low melanosome density in previous specimens (6, 7, 12). They were covered instead by an amorphous organic matrix (fig. S1), suggesting the absence of visible melanosomes may be preservational and not representative of melanosome absence and white (or other pigmentary) coloration.

Evidence of a sheetlike, end-to-end orientation of melanosomes (Fig. 2F) was present in 10 samples from BMNHC PH881. Thickness of the keratin cortex overlaying the melanosomes in barbules determines the color (e.g., black-blue sheen) of many glossy black iridescent feathers (3), but the lack of preserved keratin prevented the assignment to a particular iridescent hue. Co-presence of other pigment types like carotenoids

Fig. 1. *Microraptor* specimen BMNHC PH881. (A) Photographic image and (B) color-coded image (blue-gray, bone; orange, feathers). Numbers in (B) indicate location of samples used in assessment of preserved melanosomes and reference table S2.



in *Microraptor* cannot be excluded but would be masked by the dense melanin deposition. These data allow a conservative reconstruction of *Microraptor* as black with a glossy, weakly iridescent sheen (Fig. 4). Interpretations of *Microraptor* as nocturnal based on scleral ring morphology (13) contrast with its dark glossy plumage, a trait not found in extant nocturnal birds (14).
Generation of iridescent color by simple arrays of elongate, high-aspect ratio melanosomes, shared by extant birds and *Microraptor*, may be

part of the color repertoire of a Middle Jurassic paravian ancestor. However, given known homoplasy in Aves and the phylogenetic lability of iridescence in particular (4, 15), multiple origins of iridescence in early Paraves must be considered. Indeed, a sufficiently high density of elongate melanosomes may trigger physical processes that lead to self-assembly into iridescent nanostructures during barbule development (16), suggesting that iridescence could evolve easily relative to other complex traits. Hypotheses concerning the

function of iridescent colors have centered on intraspecific communication (17). Many are used in courtship displays (17, 18), and studies have shown that female birds prefer males with brighter iridescent color (19) and that its expression is condition-dependent (20, 21), a key prediction of honest advertisement models of sexual selection (22). Indeed, although it has only recently been studied as a distinctive color type, several extant bird species are sexually dimorphic in glossy color (23). Other functions like

Fig. 2. Scanning electron micrograph images of representative melanosome samples from (A) brown, (B) “penguin-type” brown-black [a distinct melanosome morphology; see (12)], (C) gray, (D) black, (E) iridescent extant avian feathers, and (F) BMNHC PH881. Insets of the sampled species are from (14): tufted titmouse, *Baeolophus bicolor*; macaroni penguin, *Eudyptes chrysolophus*; double-crested cormorant, *Phalacrocorax auritus*; palm cockatoo, *Probosciger aterrimus*; Brazil duck, *Amazonetta brasiliensis*; dot illustrates approximate location of sampling. (Bottom) Quadratic discriminant analysis of measured melanosome properties (8) in 168 samples of black, brown, gray, extant penguin brown-black (blue), iridescent (purple), and BMNHC PH881 feathers. Numbers refer to BMNHC PH881 samples (Fig. 2 and table S2), and m refers to a Grube Messel feather with iridescent nanostructure (5). For simplicity, only the first two axes (explaining 88% of the variance; see SOM) are shown. Circles indicate 95% confidence limits of means.

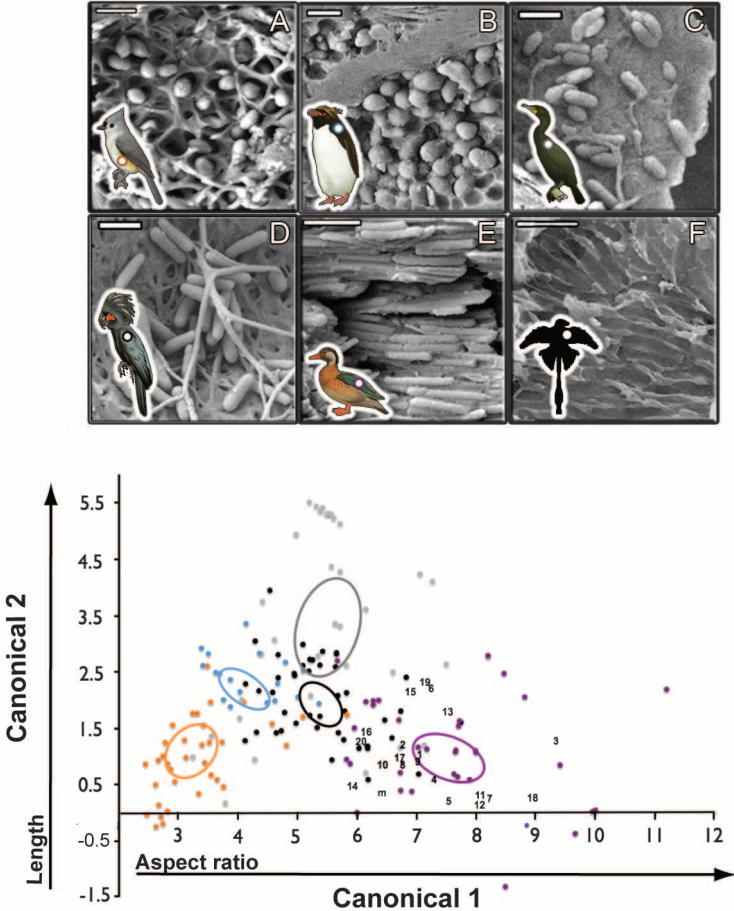


Fig. 3. Melanosome characteristics of extant feather samples and BMNHC PH881. Means ± 1 SE and one-way analysis of variance (ANOVA) of the measurements taken from melanosomes in feathers of extant bird species and BMNHC PH881. Values in the same row not sharing the same letter in superscript are significantly ($P < 0.05$) different from one another (Student’s t test). Melanosomes in column labeled “Penguins” are sampled from crown Sphenisciformes (Spheniscidae), which have a morphology (~low aspect ratio, large size) distinct from other birds (12).

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Fig. 4. Reconstruction of the Early Cretaceous paravian dinosaur, *Microraptor*: Inferred color has its basis in analysis of BMNH PH881, whereas assessment of plumage also incorporated data from digital overlays of the plumage in eight previously published specimens (8).

camouflage, interspecific signaling, and feather strengthening through melanosome alignment have been proposed but are largely untested (16). Although we cannot assign a definitive function to iridescence in *Microraptor*, a role in signaling aligns with data on the plumage of BMNH PH881.

BMNH PH881 has a complete set of distal tail feathers preserved (Fig. 1 and fig. S5). The longest tail feathers in *Microraptor* are a midline pair that form the most posterior part of the fan (Figs. 1 and 4 and figs. S5 and S6). Previously interpreted as a relatively broad, teardrop-shaped surface (9, 24) or as paired, laterally expanded rectangular winglike surfaces (25) with an important contribution to lift and stability (24, 25), the fan surface area (e.g., linked to lift production) is appreciably narrower. This reconstruction is consistent with evidence from two other specimens with well-preserved tail feathering (8) (fig. S6). Although reassessment here of *Microraptor* feathering based on BMNH PH881 and eight

previously published specimens (Fig. 4) (8) will provide a basis for further model-based estimation of potential aerodynamic implications, the graduated tail shape with a pair of long, midline feathers may be consistent with an ornamental or signaling function (26).

Paired single, or several, sets of elongate streamer-like tail feathers are seen in basal birds, including all known Enantiornithes and *Confuciusornis*, and have been interpreted as sexual ornaments (27, 28). A similar elongate streamer morphology (29) as well as patterned tails with stripes and spangles (7) are now known in several non-avian maniraptorans. Tail feathering may have commonly functioned as an ornament in coelurosaurs despite considerable modification in bony tail form and presumably function. The long bony tail in Paraves is inherited from nonflying ancestors (30). It has been ascribed utility in basal paravian maneuvering, lift generation, and landing, but it is also lost early in the evolution of birds. Tail-feather fanning musculature or nov-

el functional linkage with the forelimb has been inferred only later in bird evolution in short-tailed taxa (27, 30). Although tail structure and locomotor role shifted substantially, a signaling function may have been maintained. Iridescent color and the plumage in BMNH PH881, with other recent findings (6, 7, 27, 29), suggest that feather ornaments were diverse and that multiple cues may have been copresent (31). Sexual selection [e.g., as honest signals (21), arbitrary ornaments, or congener recognition (32, 33)] may be expected to interplay importantly with selection for aerodynamic attributes early in the evolution of birds.

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specimen is accessioned to the Beijing Museum of Natural History (BMNH). Data are presented in the supporting online material.

Supporting Online Material

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Materials and Methods

Figs. S1 to S9
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Intensifying Weathering and Land Use in Iron Age Central Africa

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About 3000 years ago, a major vegetation change occurred in Central Africa, when rainforest trees were abruptly replaced by savannas. Up to this point, the consensus of the scientific community has been that the forest disturbance was caused by climate change. We show here that chemical weathering in Central Africa, reconstructed from geochemical analyses of a marine sediment core, intensified abruptly at the same period, departing substantially from the long-term weathering fluctuations related to the Late Quaternary climate. Evidence that this weathering event was also contemporaneous with the migration of Bantu-speaking farmers across Central Africa suggests that human land-use intensification at that time had already made a major impact on the rainforest.

A major vegetation change occurred in Central Africa during the third millennium before the present (B.P.), when mature evergreen trees were abruptly replaced by savannas and secondary grasslands (1–4). The consensus is that the forest disturbance was caused by a regional climate change (1–4). However, this episode of forest clearance occurred contemporaneously with the migration of Bantu-speaking peoples from near the modern-day Nigeria-Cameroon border (5–9). The so-called Bantu expansion led to the diffusion of agriculture and iron-smelting technology across Central Africa, with potential impacts on the environment (10). Whether the Bantu farmers played an active role in the Central African deforestation event remains an open question.

To provide further constraints on this issue, we used a marine sediment record recovered off the mouth of the Congo River to reconstruct the late Quaternary history of chemical weathering in Central Africa (Fig. 1). This core (KZAI-01; 05°42'S, 11°14'E), collected at a water depth of 914 m, provides a continuous record of the Congo River sediment discharge for roughly the past 40,000 years [see supporting online material (SOM)]. Although changes in chemical weathering intensity on continents are driven primarily by natural factors—such as physical weathering rates, vegetation, rainfall, and temperature (11, 12)—intensive land use and accelerated soil denudation by increasing the surface area of minerals

and rocks exposed to weathering can also dramatically lead to much higher rates of chemical alteration (13). The degree of chemical weathering of fine-grained sediments can be inferred from the ratio of aluminum to potassium (Al/K). Potassium is highly mobile during chemical weathering and typically depleted in soils, whereas aluminum is one of the most immobile elements, being incorporated into secondary clay minerals such as kaolinite (see SOM text). High Al/K ratios in Congo fan sediments are therefore considered to be indicative of periods of intense chemical weathering in the Congo basin (14). Because downcore variations of the bulk chemical composition can also reflect changes in the sediment source, we measured neodymium (Nd) and hafnium (Hf) isotopic ratios to discriminate between both weathering and provenance signals in our sediment record. The Nd isotopic signature of terrigenous sediments is retained during continental weathering and subsequent transport, thereby providing direct information on the geographical provenance of sediment (15). Hafnium isotopes exhibit globally similar behavior but are also prone to substantial fractionation during chemical weathering, because incongruent dissolution of silicate rocks leads to products of erosion having very distinctive but systematic Hf isotopic signatures (see SOM text) (16, 17).

In this study, we quantitatively determined the bulk sedimentary major element composition of KZAI-01 at a 5-cm sampling interval (Fig. 2), corresponding to a temporal resolution of ~100 to 400 years. The age model for KZAI-01 is based on accelerator mass spectrometry (AMS) radiocarbon measurements of mixed marine carbonate fractions and tuning to a well-dated nearby sediment record from the Gulf of Guinea (GeoB6518-1;

see location in Fig. 1). We also obtained additional age constraints from ¹⁴C-AMS dating of bulk sediment organic carbon (Fig. 2). Figure 3A shows that the Nd isotopic composition of sediments deposited at site KZAI-01 (average $\epsilon_{\text{Nd}} \sim -15.9 \pm 0.6$) is almost constant and very similar to that reported for present-day riverine particulates from the Congo basin (18). This indicates that the source of material delivered to the ocean by the Congo River has remained unchanged during the Late Quaternary. In contrast, Hf isotopes display significant downcore variations (from $\epsilon_{\text{Hf}} \sim -6.8$ to -13.9), which correlate well with the Al/K depth profile (Fig. 3, A and B). Because grain size is homogeneous in this core, with medians ranging from 4 to 6 μm (17), the large range of ϵ_{Hf} values cannot be explained by changes in the relative proportions of mineral phases with distinct Hf isotope signatures. Therefore, these data show that downcore fluctuations of both ϵ_{Hf} and Al/K ratios at site KZAI-01 reflect variations in chemical weathering intensity within the Congo River drainage basin, rather than changes in sediment provenance and/or grain size.

Comparison of our proxy data with organic geochemical and molecular records from core GeoB6518-1 suggests that much of the weathering signal at site KZAI-01 is driven by continental precipitation. From ~20,000 to 3500 years ago, our weathering record exhibits strong correlation with the precipitation signal from core GeoB6518-1, inferred from the deuterium composition of plant waxes (Fig. 3C) (19, 20). This observation suggests that chemical alteration in the Congo basin has responded quickly to regional climatic changes, at least for the time scales being considered here. The trends toward wetter conditions that are visible in the GeoB6518-1 deuterium record, between ~18 to 13 and ~12 to 9 ky B.P., coincide well with marked periods of intensifying chemical weathering. Similarly, the progressive onset of dryer conditions since ~6 ky B.P., which marks the end of the African Humid Period, is accompanied by lower weathering rates. Reduced weathering rates also occurred during the Younger Dryas, between ~12.8 and 11.5 ky B.P., a period characterized by lower precipitation levels in Central Africa (19). In comparison, the evolution of mean annual temperatures in Central Africa has been very gradual since the last deglaciation, rising smoothly from ~21° to 25°C (21). Most likely, this suggests that temperature only played a minor role in controlling past chemical weathering variations in the Congo basin during the Late Quaternary.

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